

केन्द्रीय विद्यालय संगठन, कोलकाता संभाग
KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION

प्री- बोर्ड परीक्षा / PRE-BOARD EXAMINATION – 2024-25

कक्षा / CLASS – XII
विषय/SUB. - PHYSICS

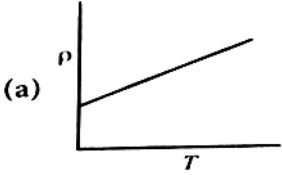
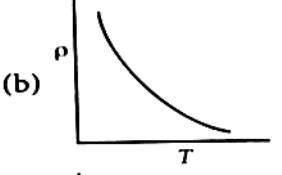
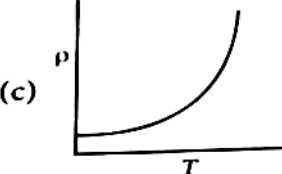
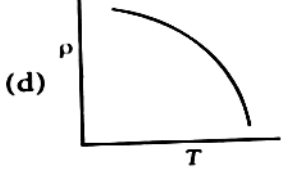
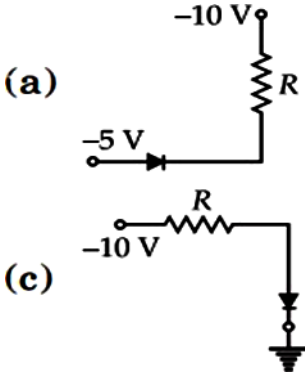
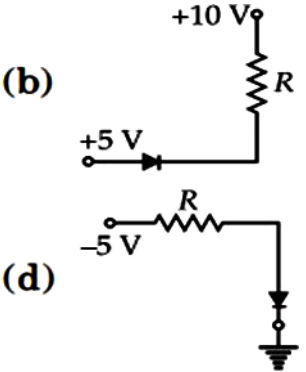
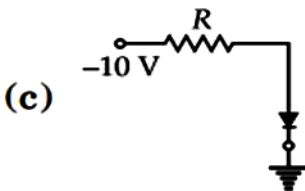
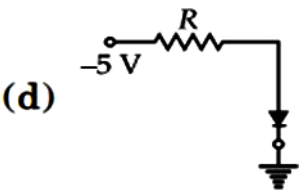
अधिकतम अंक/MAX. MARKS – 70
समय/TIME – 03 घंटे/Hours

General Instructions

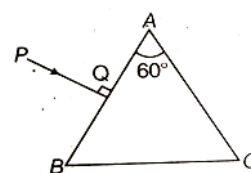
- (1) There are 33 questions in all. All questions are compulsory.
- (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- (3) All the sections are compulsory.
- (4) Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study-based questions of four marks each and Section E contains three long answer questions of five marks each.
- (5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- (6) Use of calculators is not allowed.
- (7) You may use the following values of physical constants where ever necessary
 - i. $c = 3 \times 10^8 \text{ m/s}$
 - ii. $m_e = 9.1 \times 10^{-31} \text{ kg}$
 - iii. $m_p = 1.7 \times 10^{-27} \text{ kg}$
 - iv. $e = 1.6 \times 10^{-19} \text{ C}$
 - v. $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
 - vi. $h = 6.63 \times 10^{-34} \text{ J s}$
 - vii. $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
 - viii. Avogadro's number = 6.023×10^{23} per gram mole

SECTION – A

1	The potential energy of an electric dipole placed in a uniform electric field is minimum when the dipole is: (a) Aligned with the field (b) Opposed to the field (c) Makes an angle of 45° with the field (d) Makes an angle of 90° with the field	1
2	A charge q is located at the centre of a cube. the electric flux through any two consecutive faces will be : (a) $\frac{\pi q}{6(4\pi\epsilon_0)}$ (b) $\frac{q}{6(4\pi\epsilon_0)}$ (c) $\frac{2\pi q}{6(4\pi\epsilon_0)}$ (d) $\frac{4\pi q}{3(4\pi\epsilon_0)}$	1
3	Four charges $+8Q, -3Q, +5Q$ and $-10Q$ are kept inside a closed surface. What will be the outgoing flux through the surface. (a) 26 V-m (b) 0 V-m (c) 10 V-m (d) 8 V-m	1

4	In Nuclear reaction, there is conservation of (a) Mass only (b) energy only (c) momentum only (d) mass, energy and momentum	1
5	Which of the following graphs represents the variation of resistivity ρ with temperature T for copper?  (a)  (b)  (c)  (d)	1
6	Which of the junction diodes shown below are forward biased?  (a)  (b)  (c)  (d)	1
7	A bulb is connected in series with an inductor and an alternating voltage source. The switch is turned on and the bulb glows. If now an iron rod is inserted in the inductor. The brightness of the bulb will (a) Increase (b) decrease (c) remain same (d) cannot say	1
8	A charged particle is moving in a magnetic field, leaves the field without any deflection (a) The particle is positively charged (b) the magnetic field is perpendicular to the direction of motion of the particle (c) the magnetic field is parallel to the direction of motion of the particle (d) both (a) and (c)	1
9	In Young double slit Experiment with white source of light, if one slit is covered with green filter and the other with red filter, what will be observed on the screen? (a) Alternate red and green fringes (b) Alternate bright and dark fringes (c) No interference (d) There shall be separate interference pattern for green and red	1
10	Which of the following is not due to total internal reflection? (a) Working of optical fibre (b) Difference between apparent and real depth of a pond (c) Mirage on hot summer days (d) Brilliance of diamond	1
11	A ray of light strikes a transparent rectangular slab of refractive index $\sqrt{2}$ at an angle of incidence of 45° . The angle between the reflected and refracted rays is (a) 75° (b) 90° (c) 105° (d) 120°	1

12	Which of the following is used to investigate the structure of solids? (a) Gamma rays (b)infrared rays (c) X – rays (d)cosmic rays	1
Directions: (a)Both assertion and reason are correct and reason is the correct explanation of the assertion (b) Both assertion and reason are correct and reason is not the correct explanation of the assertion (c)Assertion is correct, reason is incorrect (d)Assertion and reason both incorrect		
13	Assertion: The total energy of an electron revolving in any stationary orbit is negative. Reason : Energy can have positive or negative values of energy.	1
14	Assertion: The kinetic energy of photoelectrons emitted from a metal surface increases with the increase in the frequency of the incident light. Reason: According to Einstein's photo electric equation ,the kinetic energy of photoelectrons is directly proportional to the frequency of the incident light.	1
15	Assertion: For the fission of heavy nuclei, neutrons are more effective than protons. Reason: Neutrons are heavier than protons .	1
16	Assertion: The magnetic field of an electromagnet can be increased by increasing the number of turns. Reason: The magnetic field is directly proportional to the number of turns.	1
SECTION – B		
17	Define mobility of a charge carrier .Write the relation expressing mobility in terms of relaxation time? Give it's S.I unit.	2
18	A galvanometer coil of 50Ω resistance shows full scale deflection for a current of 5mA.How will you convert this galvanometer into a voltmeter of range 0 to 15 V?	2
19	The ratio of the intensities at minima to the maxima in the Young's double slit experiment is 9:25.Find the ratio of the widths of the two slits? OR Laser light of wavelength 630nm incident on a pair of slits produces an interference pattern in which bright fringes are separated by 7.2mm.Calculate the wavelength of another source of laser light which produces interference and bright fringes separated by 8.1mm using same pair of slits.	2
20	(i)Define the term stopping potential. (ii)Plot a graph showing the variation of photoelectric current as a function of anode potential for two light beams of same intensity but of different frequencies ν_1 and ν_2 ($\nu_2 > \nu_1$)	2
21	Draw a plot of binding energy per nucleon (BE/A)versus mass number(A) for a large number of nuclei lying between $2 < A < 240$. Using this graph , explain clearly how the energy is released in both the process of nuclear fission and nuclear fusion.	2
SECTION – C		
22	(i)Draw a ray diagram for the formation of image by a Cassegrain telescope (ii)Why these types of telescope are preferred over refracting type telescopes.	3
23	Draw the circuit diagram of a full wave rectifier using P – N junction diode. Explain its working and show the output and input waveforms.	3
24	A ray PQ is incident normally on the face AB of a triangular prism of angle of 60° made of a transparent material of refractive index $\frac{2}{\sqrt{3}}$ as shown in figure .Trace the path of the ray as it passes through the prism .Also calculate the angle of emergence and angle of deviation.	3

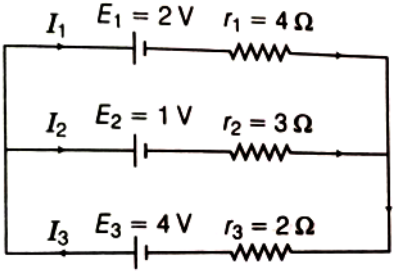
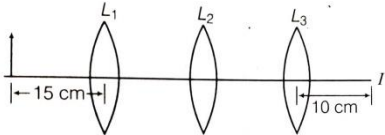


25	State Gauss's theorem in electrostatics .Using this theorem, derive an expression for the electric field due to an infinitely long straight wire of linear charge density λ . OR (i)Define electric flux and write its S.I unit. (ii)Use Gauss's law to obtain the expression for the electric field due to a uniformly charged infinite plane sheet .	3
26	(i)Explain with the help of a diagram the formation of depletion region and barrier potential in p – n junction. (ii)How is forward biasing different from reverse biasing in a p-n junction diode(only 2 points)	3
27	A long straight wire of a circular cross-section of radius a carries a steady current I .The current is uniformly distributed across the cross –section .Apply Ampere's circuital law to calculate the magnetic field at a point in the region for (i) $r < a$ (ii) $r > a$	3
28	Find the expression for the capacitance of parallel plate capacitor of plate area A and plate separation d when (i)a dielectric slab of thickness t and (ii) a metallic slab of thickness t ,where ($t < d$)are introduced one by one between the plates of the capacitor .In which case would the capacitance be more and why?	3

SECTION – D

29	When a charged particle q is moving with velocity $\vec{v}$ in a region in which electric field $\vec{E}$ and magnetic field $\vec{B}$, both are present, it experiences a force due to both the fields simultaneously. This force is known as Lorentz force. When $\vec{v}$, $\vec{E}$ and $\vec{B}$ all are collinear, the particle will follow a straight – line path with change in speed. When $\vec{v}$, $\vec{E}$ and $\vec{B}$ are mutually perpendicular, the particle will pass through the field with same velocity without any deviation in its path. (i) The magnetic Lorentz force experienced by a charge q , entering a magnetic field $\vec{B}$ with a velocity $\vec{v}$ is (a) $\sqrt{\vec{B} \times \vec{v}}$ (b) $q(\vec{B} \times \vec{v})$ (c) $q(\vec{v} \cdot \vec{B})$ (d) $(\vec{v} \times \vec{B})$. (ii) In a region, uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in mid region. The path of the particle will be (a) circle (b) ellipse (c) helix (d) straight line (iii) An electron travelling with velocity $\vec{v}$, enters a region of space in which electric and magnetic fields exist. Then the electron goes undeflected for all values of fields (a) if both electric and magnetic fields are normal to $\vec{v}$. (b) if the magnetic field alone is normal to $\vec{v}$. (c) if both electric and magnetic fields are parallel to $\vec{v}$. (d) if the electric field alone is normal to $\vec{v}$. (iv) An ion with a charge of $+3.2 \times 10^{-19}\text{C}$ is in a region where a uniform electric field of $5 \times 10^4 \text{ V/m}$ is perpendicular to a uniform magnetic field of 0.8 T. If its acceleration is zero, then its speed must be (a) 0 (b) $1.6 \times 10^4 \text{ m/s}$ (c) $4.0 \times 10^4 \text{ m/s}$ (d) $6.3 \times 10^4 \text{ m/s}$ OR The direction of magnetic force can be determined by (a)Maxwell's right hand thumb rule (b)Fleming's left hand rule (c) Maxwell's left palm rule (d) Fleming's right hand rule	4
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30	Photo electric effect: It is the phenomenon of emission of electrons from a metallic surface when light of a suitable frequency is incident on it. The emitted electrons are called photoelectrons. Nearly all metals exhibit these effect with ultraviolet light but alkali metals like Lithium, sodium, potassium, cesium ,etc .show this effect even with visible light. It is an instantaneous process i.e photoelectrons are emitted as soon as the light is incident on the metal surface .The number of photoelectrons emitted per sec is directly proportional to the intensity of the incident radiation. The maximum kinetic energy of the photoelectrons emitted from a given metal surface is independent of the intensity of the incident light and depends only on the frequency of the incident light . For a given metal surface there is a certain minimum value of the frequency of the incident light below which emission of photoelectrons does not occur.	4
	(i) In a photoelectric experiment plate current is plotted against anode potential (a) A and B will have same intensities while B and C will have different frequencies (b) B and C will have different intensities while A and B will have different frequencies (c) A and B will have different intensities while B and C will have equal frequencies (d) B and C will have equal intensities while A and B will have same frequencies. (ii) Photo electrons are emitted when a zinc plate is (a) Heated (b) hammered (c) Irradiated by ultraviolet light (d) subjected to a high pressure (iii) The threshold frequency for photoelectric effect on sodium corresponds to a wavelength of 500 nm. Its work function is about (a) $4 \times 10^{-19} \text{ J}$ (b) 1J (c) $2 \times 10^{-19} \text{ J}$ (d) $3 \times 10^{-19} \text{ J}$. (iv) The maximum kinetic energy of photo electrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV. The stopping potential is (a) 2V (b) 4V (c) 6V (d) 10V OR The minimum energy required to remove an electron from a substance is called its (a) work function (b) kinetic energy (c) stopping potential (d) potential energy	
	SECTION – E	
31	(i) Derive an expression for drift velocity of electrons in a conductor .Hence ,deduce Ohm's law. (ii) A wire whose cross-sectional area is increasing linearly from one end to the other ,is connected across of V volts .Which of the following quantity or quantities remain constant in the wire ? Explain. (a) Drift speed (b) current density (c) electric field.	3+2

	OR (i) State Kirchhoff's rules. (ii) Using Kirchhoff's rules to write the expression for the currents I_1, I_2, I_3 in the circuit shown in figure. 	
32	(i) Explain briefly, with the help of a labelled diagram, the basic principle of the working of an a.c.generator. (ii) In an a.c. generator, coil of N turns and area A is rotated at an angular velocity ω in a uniform magnetic field B. Derive an expression for the instantaneous value of the emf induced in coil. (iii) What is the source of energy generation in this device? OR (i) With the help of a diagram, explain the principle of a device which changes a low ac voltage into a high voltage. Deduce the expression for the ratio of secondary voltage to the primary voltage in terms of the ratio of the number of turns of primary and secondary winding. For an ideal transformer, obtain the ratio of primary and secondary currents in terms of the ratio of the voltages in the secondary and primary coils. (ii) Write any two sources of the energy losses which occur in actual transformers. (iii) A step-up transformer converts a low input voltage into a high output voltage. Does it violate law of conservation of energy? Explain.	5
33.	(i) Draw a ray diagram to show the working of a compound microscope. Deduce an expression for the total magnification when the final image is formed at the near point. (ii) In a compound microscope, an object is placed at a distance of 1.5cm from the the objective of focal length 1.25cm .If the eyepiece has a focal length of 5cm and the final image is formed at the near point. Estimate the magnifying power of the microscope. OR (i) Obtain len's maker formula using the expression $\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R}$ Here, the ray of light propagating from a rarer medium of refractive index n_1 to a denser medium of refractive index n_2 is incident on the convex side of spherical refracting surface of radius of curvature R. (ii) You are given three lenses L_1, L_2, L_3 each of focal length 10cm. An object is kept at 15 cm in front of L_1 as shown. The final image I is formed at the focus of L_3. Find the separation between L_1, L_2 and L_3. 	3+2